```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy.stats import linregress
from sklearn.model_selection import train_test_split
from sklearn.linear_model import LinearRegression
from sklearn import metrics
```

```
df = pd.read_csv("RegData.csv")
```

```
df.sample(5)
```

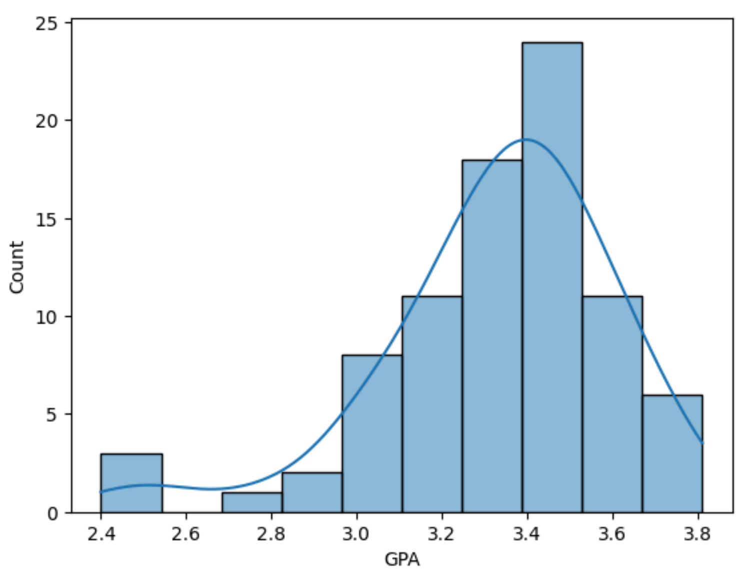
	StID	SAT	GPA
56	168	1134	3.42
71	185	1278	3.59
0	101	1355	3.42
78	192	1202	3.39
48	159	1466	3.38

```
import warnings
warnings.filterwarnings("ignore")

def dist2Chart(c2):
  plt.figure(figsize=(16,5))
  plt.subplot(1,2,2)
  sns.distplot(df[c2])
  plt.show()
```

```
sns.histplot(data=df, x="GPA", kde=True)

plt.show()
```




```
df.head()
```

	StID	SAT	GPA
0	101	1355	3.42
1	102	1391	3.48
2	103	1170	2.91
3	104	1357	3.41
4	105	1326	3.28

```
df.shape
```

```
(84, 3)
```

```
df["GPA"].quantile(0.25)
df["GPA"].quantile(0.5)
df["GPA"].quantile(0.75)
```

```
np.float64(3.5025)
```

```
df["GPA"].median()
```

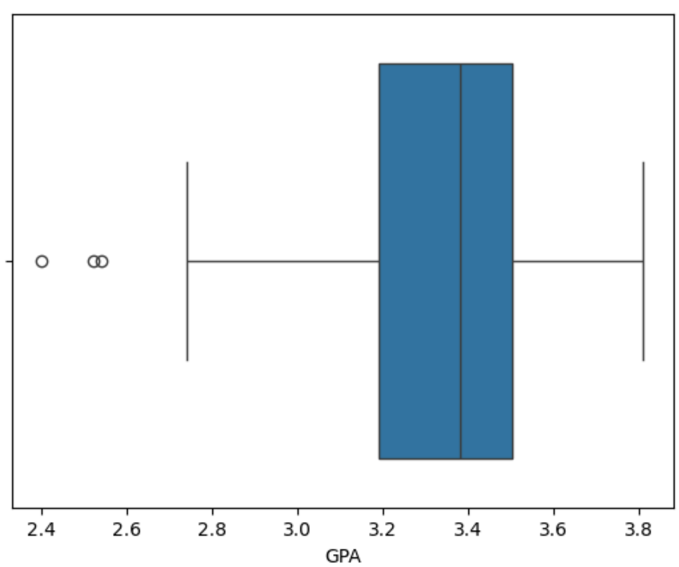
```
3.38
```

```
# IQR method
def outlier_remover(df, column):
  q1 = df[column].quantile(0.25)
  q3 = df[column].quantile(0.75)
  IQR = q3-q1
  lower_bound = q1-1.5*IQR
  upper_bound = q3+1.5*IQR
  df = df[(df[column] > lower_bound) & (df[column] < upper_bound)]
  return df
```

```
df1 = df.copy()
```

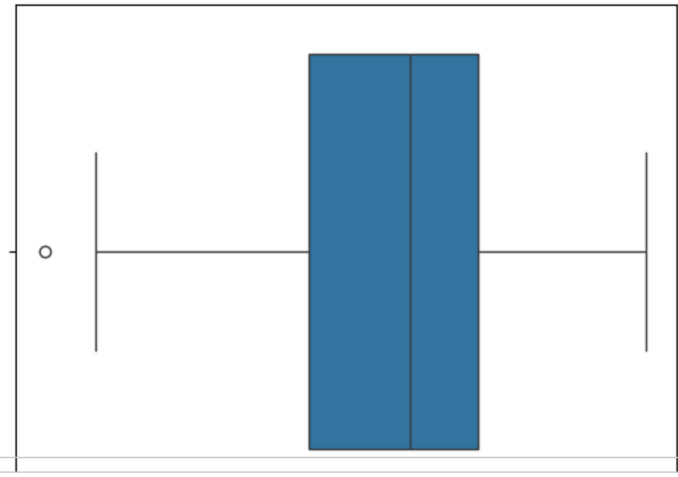
```
sns.boxplot(data=df, x=df["GPA"])
```

```
<Axes: xlabel='GPA'>
```




```
df1 = outlier_remover(df1, "GPA")
sns.boxplot(data=df1, x=df1["GPA"])
```

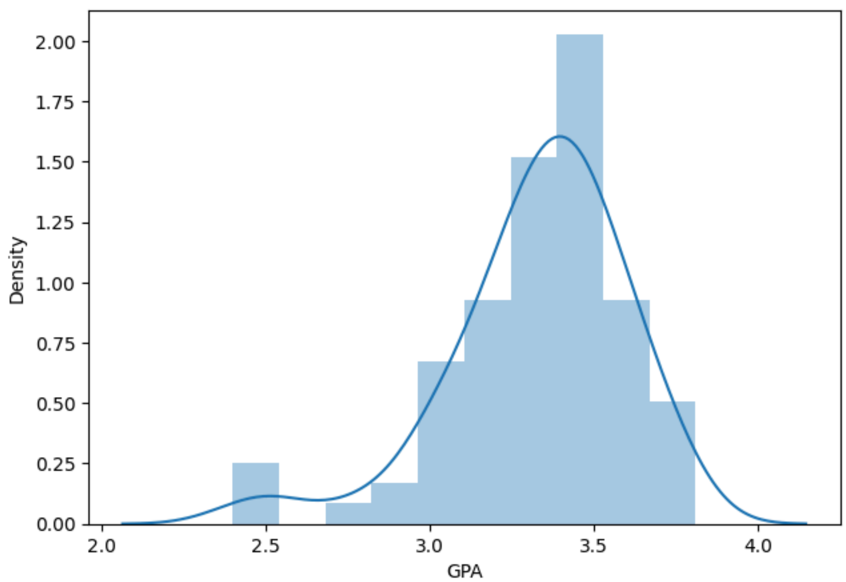
```
<Axes: xlabel='GPA'>
```




```
df1.shape
```

```
(81, 3)
```

```
dist2Chart("GPA")
```




```
df1.to_csv("RegData_new.csv")
```

```
df1.shape
```

```
(81, 3)
```

Start coding or generate with AI.